MATHEMATICS ALIVE – INFORMATION PROCESSING IN REAL LIFE



Biological example for the Fibonacci numbers



Mason packing bricks for constructing a wall between two pillars

7.2 Principles of Counting

There are some basic counting techniques which will be useful in determining the number through different ways of arranging or selecting objects. The basic counting principles are given below.

7.2.1 Addition principle

If there are two selections such that they can be done **independently** in **m** ways and **n** ways respectively, then either of the two selections can be done in (**m** + **n**) ways.

Let us learn about this addition principle of counting as given below :

Situation:

In class VIII, there are 16 boys and 9 girls. The teacher wants to select either a boy or a girl as the class leader. Let us see, in how many ways can the teacher select the class leader.



Fig. 7.1

The teacher can select the class leader in any one of the following ways.

(i) In the first choice, the teacher can select **a boy among 16 boys in 16 ways** (who ever may be of the 16 boys).

(ii) In the second choice, the teacher can select **a girl among 9 girls in 9 ways** (who ever may be of the 9 girls).

Hence, the teacher can select the class leader who is **a boy** or **a girl** in **25 different ways (16 boys + 9 girls).**

Thus, we come to know, if a selection **A** can occur in ***m*** ways and another selection **B** can occur in ***n*** ways, and suppose that **both cannot occur together**, then **A** or **B** can occur in (***m*** + ***n***) ways. Let us see an example.

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